Assignment 2
Ordinary Differential Equations

Group 19
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PART 1: Fast and slow adsorption (linear system)

Exercise 1: Define and program the two systems, then verify and test your solution

a) Reversible equations:

$$\frac{dm_D}{dt} = -k_{DA}m_D + k_{AD}m_A$$

$$\frac{dm_A}{dt} = -k_{AS}m_A - k_{AD}m_A + k_{DA}m_D + k_{SA}m_S$$

$$\frac{dm_S}{dt} = -k_{SA}m_S + k_{AS}m_A$$

Irreversible equations:

$$\frac{dm_D}{dt} = -k_{DA}m_D$$

$$\frac{dm_A}{dt} = -k_{AS}m_A + k_{DA}m_D$$

$$\frac{dm_S}{dt} = k_{AS}m_A$$

b) t= 12 d: m_D= 0.219519046 g; m_A= 0.423049 g; m_S= 0.357432 g

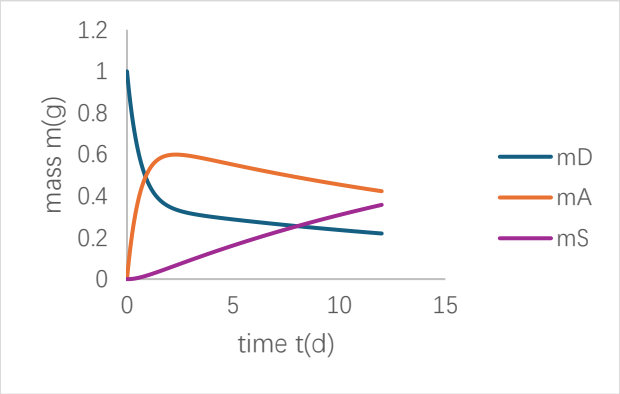


Figure 1 Mass-time Variation Curve of Reversible Process

c) t= 12 d: m_D= 0.00000323 g; m_A= 0.49881094 g; m_S= 0.50118583 g

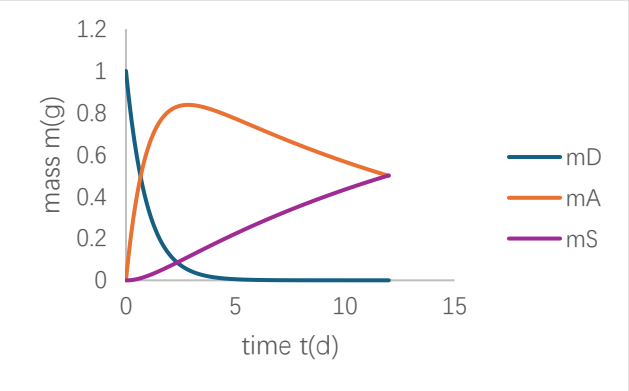


Figure 2 Mass-time Variation Curve of Irreversible Process

d) t = ∞:

$$\frac{dm_D}{dt} = -k_{DA}m_D + k_{AD}m_A = 0;$$

$$\frac{dm_A}{dt} = -k_{AS}m_A - k_{AD}m_A + k_{DA}m_D + k_{SA}m_S = 0;$$

$$\frac{dm_S}{dt} = -k_{SA}m_S + k_{AS}m_A = 0.$$

Rearranging : $\frac{C_A}{C_D} = \frac{m_A}{m_D} = \frac{k_{DA}}{k_{AD}} = \frac{1.00}{0.50} = 2$; $\frac{C_S}{C_A} = \frac{m_S}{m_A} = \frac{k_{AS}}{k_{SA}} = \frac{0.0632}{0.01} = 6.32$

Exercise 2: Compare analytical and numerical solutions

Table 1 Result of Exercise 2, mass in D,A and S

	D	A	S	Student
Analytical solution	0.000006144	0.500009765	0.499984091	Jiayue Che Qiaowen You
Excel dt=2d	1.000000000	-0.592971733	0.592971733	
Excel dt=0.003d	0.000006034	0.499973930	0.500020036	
Python best solver [Radau]	0.000006152	0.500009759	0.499984093	Qiaowen You Yining Gan
Python worst solver [RK23]	0.000003426	0.500010731	0.499985843	

Table 2 Relative deviation (numerical result divided by outcome of analytical solution)

	D	A	S	Student
Analytical solution	1.00000000	1.00000000	1.00000000	Jiayue Che Qiaowen You
Excel dt=2d	162754.79141900	-1.18592030	1.18598120	
Excel dt=0.003d	0.98212560	0.99992833	1.00007189	
Python best solver [Radau]	1.00126748	0.99999999	1.00000000	Qiaowen You Yining Gan
Python worst solver [RK23]	0.55759792	1.00000193	1.00000350	

The 2d time step is too high, leading to result oscillations.

PART 2: Unified model for sorption and biodegradation

Exercise 3: Implement and test bacterial growth model / Plausibility check

To obtain the dissolved substrate concentration when the bacterial growth rate is 0 d^{-1}, solve for the case where eq.2 is equal to zero.

$$\frac{dX}{dt} = \frac{\mu_{max}C_D}{K_M + C_D} \times X - b \times X = 0$$

And then we can obtain the following equation through transposing terms.

$$\frac{\mu_{max}C_D}{K_M + C_D} \times X = b \times X$$

Since the biomass of bacteria cannot be zero is a reaction, otherwise the reaction cannot proceed, X≠0. We can cancel out X on both sides, rearrange the equation, and substitute the known data ($\mu_{max} = Y \times v_{max}$):

$$C_D = \frac{b \times K_M}{\mu_{max} - b} = \frac{0.05\ d^{-1} \times 0.715\ g/m^3}{0.31 \times 4 - 0.05\ d^{-1}} = 0.030\ g/m^3$$

That is, when C_D=0.030g/m^3, the growth of bacteria $\frac{dX}{dt} = 0\ d^{-1}$.

The way to check the numerical solutions for non-linear differential equation system:

1. Verify the reliability of the calculation results in special scenarios, such as whether the substrate concentration value is consistent with the result of solving the single equation when the growth rate of bacteria is zero. At the same time, it is essential to observe whether numerical solutions satisfy physical laws, such as mass conservation or boundary constraints.

2. Reduce the step size to half of the one to be verified and use the corresponding solutions of variables to calculate the error ratios. If their values are less than the selected threshold (1% is selected this time), the Euler method is implemented correctly. We use this method to test our numerical solutions (Table 3). All the values of error ratio are under 1%, so we preliminarily justify the calculation result is correct.

$$error\ ratio = \frac{|y_{\Delta t/2} - y_{\Delta t}|}{y_{\Delta t/2}}$$

Table 3 numerical solution test for dt=0.1d at t=60d

dt	m_D	m_A	m_S	m_M	X	CO_2
0.1	0.007476004	0.01728702	0.242766337	0.732470639	0.025591865	0.609641757
0.05	0.007427571	0.017174557	0.241162998	0.734234875	0.02558416	0.611136389
error ratio	0.65%	0.65%	0.66%	0.24%	0.03%	0.24%

3. When comparing the calculation results of Euler method and python's 'solve_ivp' methods, if the curves of different algorithms almost overlap, the numerical solutions are reliable.

Exercise 4: Simulations with microbe

a)

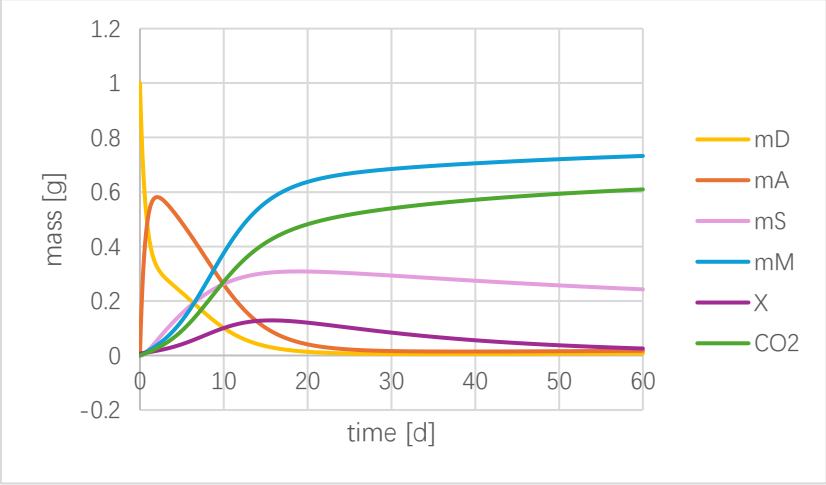


Figure 3 Mass-time Variation Curve of Reversible Process

b) t = 60 d:

m_D= 0.007476004 g; m_A= 0.017287 g; m_S= 0.2427663 g; m_X= 0.025591865 g; m_{CO2}= 0.609641757 g.

c) D: Dissolved 2,4-D; A: Adsorbed 2,4-D; S: Sequestered 2,4-D;

X: Carbon in living microbial biomass (not 2,4-D); CO_2: Mineralized product (not 2,4-D).

d) Identify the peak (Excel): X (0.128868425 g) maximizes at t=15.7 d.

e) From my Excel simulation, the specific growth rate dX/(Xdt) decreases continuously over time (due to substrate depletion). Thus, the maximum observed occurs at t = 0 (initial time), with $dX/(Xdt)_{max}$= 0.6730321 d^{-1}.

$\mu_{max} = Y \cdot v_{max}$ = 0.31*4 d^{-1}= 1.24 d^{-1}, μ_{max} is theoretical, the simulated value is smaller because substrate (m_D) is not infinite (limited by Michaelis-Menten kinetics).

Exercise 5: Close the mass balance

a) m_D= 0.007476004 g; m_A= 0.017287 g; m_S= 0.2427663 g; m_X= 0.025591865 g; m_{CO2}= 0.609641757 g

sum up: m(14C)= 0.902762984 g

b) The missing labeled carbon is due to the fact that after microbial death(decaying X), it is converted into bioNER.

c)

$$\frac{dm(dead\ X - bioNER)}{dt} = f \times b \times X$$

Substitute the data, $m(dead\ X - bioNER)$= 0.104237016 g

The final amount is 1.007 g, and 0.007 g is the initial microbial degrader biomas X. Hence, the final amount of labeled carbon (C) is 1 g, and it is the same as the initial 14C in compartment D.

Exercise 6: Sensitivity Analysis

a) In Exercise 1(d), the equilibrium ratios were $C_A/C_D = 2.0$ and $C_S/C_A = 6.32$. At t = 60 d, the ratios are not valid anymore since biodegradation depletes D and prevents equilibrium.

b) The final mass of ^{14}C in biomass for X at t=60 changed from 0.02559 g to 0.02408 g. A higher initial microbial mass consumes substrate CD faster, leading to early saturation and growth limitation. The system is nonlinear; the final microbial mass is determined by substrate availability and death rate, not simply the initial biomass.

c) The final mass of ^{14}C in biomass for X at t=60 was 0.02592g when increasing X (0) by factor 22. The high initial microbial mass rapidly consumes substrate CD, causing an initial surge in growth, but as the substrate is depleted, growth is limited and reaches a similar equilibrium biomass. And the total mass of ^{14}C in the system was 0.95g.

d) The true mass of ^{14}C in biomass X at t=60d with X (0) =0.007 was 0.00695 g, according to equation below:

$$\frac{dX}{dt} = \mu_{max}\frac{C}{K_C + C}X - bX$$

$$\mu_{max} = Y \times v_{max}$$

e) The original parameter values were modified to observe the changes in the biomass (X)–time curve. The parameters that resulted in no microbial growth were identified, where the biomass curve appeared nearly horizontal over time. The initial biomass was set to X_0=0.007g:

Table 4 Parameter that directly affect microbial growth

Parameters	Original Value	Change Value	Changed biomass	Explanation
v_{max}	4 d^{-1}	1 d^{-1}	0.00704g	Lower v_{max} slows substrate use, limiting growth
K_m	0.715 g/m^3	5 g/m^3	0.00310g	Higher K_m reduce substrate affinity and growth
Y	0.31	0.01	0.00050g	Low Y lowers biomass yield
b	0.05 d^{-1}	0.5 d^{-1}	10^{-10}g	High b causes rapid decay
μ_{max}	1.24 d^{-1}	0.1 d^{-1}	0.00087g	Low μ_{max} prevent net growth

Exercise 7: Legal consequences

a)

Table 5 NER type of the compartment

Compartment	NER Type
D	\
A	Type I
S	Type I
CO_2	\
X	Type III
dead X	Type III

b) NER I = m_A+m_S= 0.017287 g; m_S+0.2427663 g = 0.2600533 g

NER II = 0

NER III = m_X+ m($dead\ X - bioNER$) = 0.025591865 g+ 0.104237016 g = 0.129828881 g

XenoNER = NER I + NER II = 0.2600533 g

Total NER = XenoNER + NER III = 0.389882181 g

Group Declaration

	Anqi Yu	Qiaowen You	Jiayue Che	Yining Gan
Excel code		√	√	√
Python code		√	√	√
Reading	√	√	√	√
Error search	√	√	√	√
Part 1		√ （Exercise 2）	√ (Exercise 1)	
Part 2		√ （Exercise 3）	√ (Exercise 4)	√ (Exercise 5，6)
Exercise 7	√			
Writing	√	√	√	√
Graphs	√	√	√	√
Polishing and Quality Control	√	√	√	√
Cake	√	√	√	√

Declare for Use of AI

Task List	How was Generative AI used? (1, 2, 3, 4) *	Reflections on what was learned using Generative AI
Exercise 1	1	/
Exercise 2	1	/
Exercise 3	2	How to understand the question-setting approach and search for references
Exercise 4	1	/
Exercise 5	1	/
Exercise 6	4	Polish language with AI, and learned how to write an academic report
Exercise 7	1	/

* Uses of GenAI:

1 No use of GenAI

2 GenAI used for idea generation and structuring the assignment

3 GenAI to edit language/grammar but not content

4 GenAI used to write part of the text/make figures